FEI HAN

Robot Learning · Dynamics Modeling · Bioinspired Robotics

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Joint PhD Program · Expected completion June 2027

Education

Westlake University and Zhejiang University
Joint PhD Program · Computer Science and Technology
Aug 2023 - Jun 2027 (expected) · Advisor: Prof. Dixia Fan · i4FSI Lab

University of Chinese Academy of Sciences · MEng · Sep 2019 - Jun 2022
Pattern Recognition and Intelligent Systems
Research at the State Key Laboratory of Robotics, Shenyang Institute of Automation, CAS.

Zhengzhou University · BEng
Measurement and Control Technology and Instruments · Sep 2015 - Jul 2019

Research Appointments

Pennsylvania State University · Visiting Scholar · Apr - Oct 2026

Westlake University · Research Assistant · Jul 2022 - Aug 2023

Research Contributions

Experimental data and hydrodynamic models [2, 9, 11]. Collected and processed force data, developed temporal hydrodynamic models and contributed to a real-fluid learning platform. Related work studies datasets and benchmarks for dynamic load reconstruction.

Safe learning and bioinspired locomotion [1, 3, 10]. Studied constrained policy learning, phase coordination, swimming behavior planning and policy distillation; combined air-water modeling with RL for storm-petrel locomotion.

Deep-sea soft manipulation [6-8]. Contributed to NSFC Major Research Plan Integrated Project No. 92048302, studying robots that combine rigid, flexible and soft structures for operations in extreme environments. As a core student contributor to the soft-gripper work, led design, fabrication, sensing and control development, and guided junior students under the advisor's supervision.

Multilegged swimming (Apr 2026 - Present). Ongoing reduced-order modeling and simulation with researchers from Harvard University, Carnegie Mellon University and other institutions; manuscript in preparation. | Baxi Chong group

Publications and Manuscripts

* Equal contribution. † Co-corresponding author. Applicant name in bold.

Published and Accepted

1. Xinyu Cui*, **Fei Han*†**, et al. Swimming Under Constraints: A Safe Reinforcement Learning Framework for Quadrupedal Bio-Inspired Propulsion.
ICRA · 2026 · Accepted

2. **Fei Han**, Pengming Guo, Hao Chen, et al. Learn to Swim: Data-Driven LSTM Hydrodynamic Model for Quadruped Robot Gait Optimization.
ICRA · 2025 · pp. 10261-10267 · Published

Publications and Manuscripts Continued

Published and Accepted Continued

3. Jiaqi Xue*, **Fei Han***, et al. Exploring Storm Petrel Pattering and Sea-Anchoring Using Deep Reinforcement Learning.
Bioinspiration & Biomimetics · 2023 · 18(6): 066016 · Published

4. Linlin Kang, Shiying Xiong, Buchen Wu, Chao Li, Mingming Ge, Dehan Yuan, **Fei Han**, Weicheng Cui. Force Decomposition on Flapping Flexible Plate via Impulse Theory and Dynamic Mode Decomposition.
Physics of Fluids · 2023 · 35(10): 101914 · Published

5. Linlin Kang, An-Kang Gao, **Fei Han**, et al. Propulsive Performance and Vortex Dynamics of Jellyfish-like Propulsion with Burst-and-Coast Strategy.
Physics of Fluids · 2023 · 35(9): 091904 · Published

6. **Fei Han**, Daohui Zhang, Xingang Zhao, et al. Design of a Composite-Cavity Bioinspired Soft Hand for Underwater Grasping.
Robot · 2023 · 45(2): 207-217 · Published · In Chinese; title translated

7. **Fei Han**, Daohui Zhang, Xingang Zhao. Human-Robot Interaction Control of a Soft Hand Based on a Curvature Prediction Model.
Automation & Instrumentation · 2022 · 37(7): 43-47 · Published · In Chinese; title translated

8. Yanxu Sun, **Fei Han**, Daohui Zhang, et al. An Underwater Soft Claw Based on Bionic Principle.
IEEE ROBIO · 2021 · Published

Under Review

9. **Fei Han***, Xinyu Cui*, Zhipeng Wang*, et al. Embracing Flow Unsteadiness: A High-Throughput Learning Platform Enables Vortex-Exploiting Bioinspired Propulsion.
Nature Mechanical Engineering · Under review · arXiv:2609.14254 (2026)

10. From Gaits to Routes: Execution-Aware Hybrid Behavior Planning for Quadrupedal Swimming with Experiment-Calibrated Hydrodynamics.
ICRA · 2027 · Under review

11. HydroWrench: A Dataset and Benchmark for Dynamic Load Reconstruction in Bioinspired Aquatic Mechanisms.
ICLR · 2027 · Under review

Academic Service and Campus Engagement

Westlake visits during IROS 2025. Core organizer and presenter for laboratory-hosted visits; welcomed academic and industry guests and introduced Westlake, the laboratory and its research.

Laboratory open days and science outreach. Hosted visits by primary and secondary school students and explained research platforms, bioinspired robotics and fluid mechanics.

Intelligent fluids conference. Volunteered in conference logistics and on-site service for an event hosted by the School of Engineering.

Undergraduate residential-college exchanges. Shared research experiences and reflections on postgraduate study as a graduate student.

Westlake University choir. Participated in rehearsals and university performances.